L. Las Tortuguitas

time limit per test: 1 second

memory limit per test: 1024 megabytes

Dudu is on a magical journey inside the Fantastic Chocolate Factory and discovered a secret room full of machines that produce extremely rare candies called "Tortuguitas". This room contains N machines numbered from 1 to N, and each machine produces exactly K Tortuguitas, numbered from 1 to K.

A machine is represented by the rarity of the Tortuguitas it produces. More formally, the i-th machine produces K Tortuguitas with rarities $r_{i,1}, r_{i,2}, \cdots, r_{i,K}$. Let X be the rarity value of the simplest Tortuguita, and the rarity of a Tortuguita can be calculated as follows.

- $r_{i,j} = j \cdot X$ for all $(1 \leq j \leq K)$, when $i = 1$.
- $r_{i,j} = \sum_{t=1}^{j} r_{i-1,t}$, when $i > 1$.

Dudu was amazed by these magical machines and asked for your help to answer some questions about them.

He will give you Q questions of the following type: In the i-th machine, what is the sum of the rarities of the Tortuguitas numbered greater than or equal to L and less than or equal to R?

More formally, Dudu will ask for the result of $\sum_{j=L}^{R} r_{i,j}$.

Since this result can be absurdly large, he only wants you to return the remainder of this sum when divided by 998244353.

Input

The first line of input contains 4 integers N, K, Q, and X $(1 \leq N, K, Q \leq 3 \cdot 10^5)$ $(1 \leq X < 998244353)$ — the number of machines, the number of Tortuguitas each machine produces, the number of questions Dudu asked, and the rarity of the simplest Tortuguita, respectively.

The next Q lines each contain 3 integers i, L, and R $(1 \leq i \leq N)$ $(1 \leq L \leq R \leq K)$, the machine number and the range for which Dudu wants to know the sum.

Output

The output should contain Q lines, each with the result of the sum modulo 998244353.

Examples

input

```
3 10 3 5
1 4 7
2 4 7
3 4 7
```

output

```
110
370
975
```

input

```
50 100 2 7
4 1 10
50 3 100
```

output

```
14014
574802200
```

input

```
1 1 1 1
1 1 1
```

output

```
1
```

UDESC Selection Contest 2025-1

Finished

Practice

→ About Contest

BRUTE competitive programming

Contest website

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++20 13.2 (64 bit, win

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
329339554	Jul/17/2025 16:13	Accepted
326066120	Jun/26/2025 03:04	Accepted
323683503	Jun/09/2025 22:23	Accepted

→ Contest materials

- statements (pt)
- statements (en)
- Solution Slides (pt)